1.

Use the ping command to test connectivity from your computer to www.youtube.com and note the average response time.

Ans

- Packets sent: 4
- Packets received: 4
- Packets lost: 0 (0% loss)
- Minimum response time: 9 ms
- Maximum response time: 30 ms
- **Average response time: 15 ms**

2.

through to reach the Run the traceroute (tracert on Windows) command to www.flipkart.com and list all the intermediate hops shown in the output.

Hint: Each hop represents a network device your data passes destination.

Ans

Hop	IP address	Hostname	Response time
1	192.168.1.1	gpon.net	1 ms
2	116.74.105.17	hathway.com	3–4 ms
3	116.74.100.201	hathway.com	3–11 ms
4	136.232.18.13	Not shown	9–10 ms
5	172.31.2.73	Not shown separately	18–94 ms
6	103.77.108.84	Not shown	15–16 ms
7	23.55.244.176	Akamai Technologies	13–28 ms

3.

Use the nslookup tool to find the IP address of www.zomato.com and record the result.

IP addresses found:

- IPv4: 23.55.244.185
- IPv4: 23.55.244.186
- IPv6: 2600:140f:1e00::1737:f4d0
- IPv6: 2600:140f:1e00::1737:f4c3

4.

Test connectivity to another device on your local network (such as your friend's laptop or your phone) using the ping command and describe what happens if the device is offline.

Ans 4

Command used: ping 192.168.1.10

My computer's IP address: 192.168.1.8

Target device IP address: 192.168.1.10

Result:

- **Packets sent: 4**
- **Destination host unreachable: 4 times**
- **Successful replies from the target device: 0**
- **What happens if the device is offline?** If the target device is offline or unreachable, the ping test may fail. In this test, my computer reported "Destination host unreachable."

5.

Create a short report (5-7 lines) summarizing which tool (ping, traceroute, nslookup) is most useful for diagnosing each of these problems: (a) Website not loading, (b) Slow connection, (c) Suspecting DNS issues.

Constraint: Give a specific example for each tool based on your own test results above.

Ans 5

Report: Network Troubleshooting Tools

1. **Ping:** It is useful for checking connectivity. In my test, pinging 192.168.1.10 showed “Destination host unreachable.”
2. **Traceroute (tracert):** It helps identify the network path and possible delays when a connection is slow.
3. **Nslookup:** It helps diagnose DNS issues by finding a website's IP address. My test found IPv4 addresses 23.55.244.185 and 23.55.244.186 for www.zomato.com.
4. **(a) Website not loading:** Use ping to check connectivity, then use nslookup to check DNS resolution.
5. **(b) Slow connection:** Use tracert to examine hop response times. My test to Flipkart showed 7 hops.
6. **(c) Suspecting DNS issues:** Use nslookup to check whether the domain name resolves to IP addresses.
7. **Conclusion:** Ping checks connectivity, tracert examines the network route, and nslookup checks DNS information.